ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE E · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice e.

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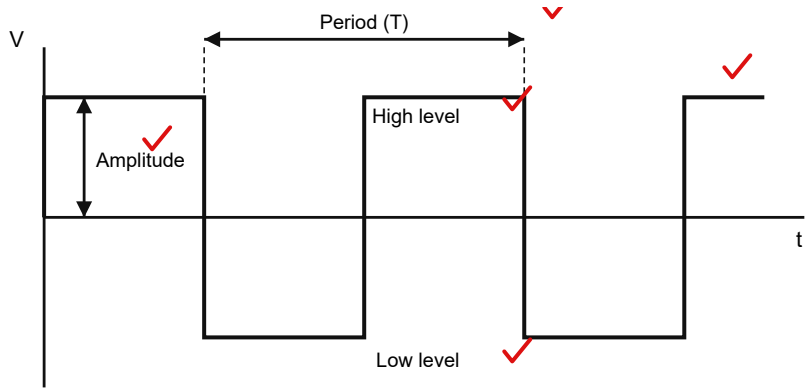
How to mark: each ✓ is one mark. Accept other correct answers and wording. Circuits may be drawn in different but correct ways: mark each required element and correct connection.

QUESTION 1

1.1 A semiconductor device with two terminals ✓ that allows current to flow in one direction only (forward bias) and blocks it in the other ✓. (2)

1.2 Anode ✓, cathode ✓, gate ✓. With the anode positive to the cathode (forward biased) ✓, a small current pulse into the gate switches the SCR on, and it stays on until the anode current drops below the holding current ✓. (5)

1.3

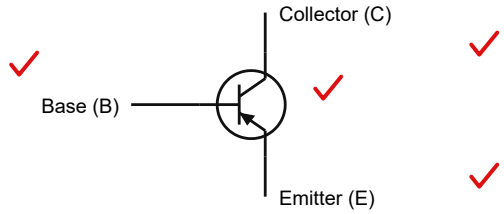


(Square wave with vertical edges and flat tops 1; high level 1 and low level 1 labelled; amplitude 1; period (one full cycle) 1)

(5)

1.4 Any TWO, ✓✓ each: a microcontroller has the CPU, memory (RAM and flash) and input/output ports on one chip, so few extra parts are needed; it is small, cheap and uses little power (good for battery robots); it has built-in ADC, PWM, timers and communication for sensors and motors; it is easy to program for one dedicated task. (4)

1.5



(Correct symbol with the emitter arrow pointing INWARDS (towards the base) 2; base 1; collector and emitter labelled 1)

(4)

[20]

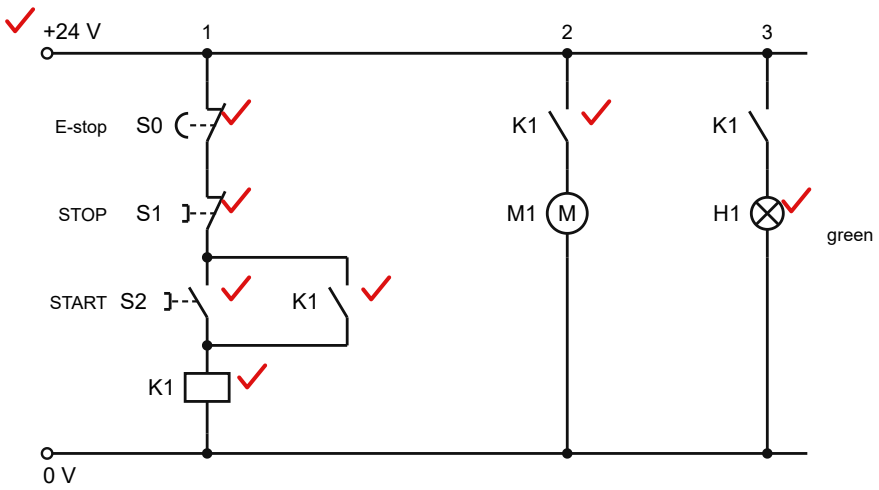
QUESTION 2

2.1 The supply is switched on and off very quickly at a fixed frequency ✓; the width of the ON pulse (duty cycle) is changed ✓; the average voltage across the motor is proportional to the duty cycle ✓, so a wider pulse gives a higher average voltage and a higher speed ✓. (4)

2.2 Controller (e.g. microcontroller / PLC giving step and direction signals) ✓; stepper driver / driver IC ✓; power supply ✓; stepper motor ✓ (also accept indexer, limit switches) (4)

2.3 Open loop: no feedback, the output is not measured or corrected ✓ – e.g. a conveyor on a timer, a stepper without an encoder ✓. Closed loop: the output is measured and fed back to correct errors ✓ – e.g. a servo motor with an encoder, temperature control of an oven ✓. (4)

2.4



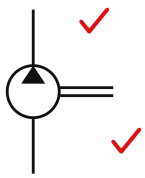
(Supply lines (+24 V and 0 V) 1; E-stop (NC) 1 and STOP (NC) 1 in series; START (NO) 1 with K1 holding contact in parallel 1; relay coil K1 1; K1 contact switching the motor 1; green lamp switched by a K1 contact 1)

(8)

[20]

QUESTION 3

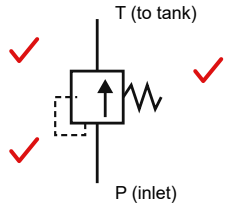
3.1.1



(Circle with ONE solid triangle pointing outwards 1, drive shaft and inlet / outlet lines 1)

(2)

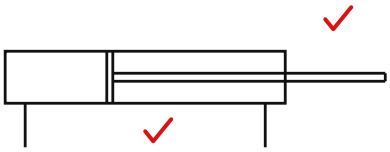
3.1.2



(Square valve with the arrow offset from the flow line (normally closed) 1; adjustable spring 1; dashed pilot line from the inlet 1)

(3)

3.1.3



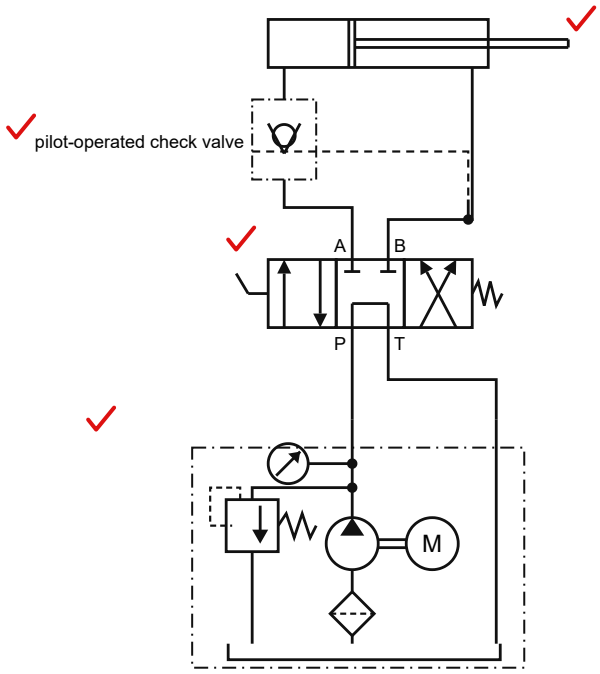
(Barrel with piston and rod 1 and TWO ports, one at each end 1)

(2)

3.2 Pressure applied to a confined liquid is transmitted equally in all directions ✓✓. (3)
Application ✓: hydraulic press, car hoist / jack, hydraulic brakes, robot or excavator arm.

3.3 Any FOUR ✓✓✓✓: transmits power; lubricates moving parts; seals clearances between parts; carries away heat (cooling); carries contaminants to the filter; protects against rust and corrosion. (4)

3.4



(Reservoir, pump with motor and filter 1; pressure relief valve from the pressure line to tank 1; 4/3 manually operated valve with a tandem centre 1; double-acting cylinder with both ports connected to A and B 1; pilot-operated check valve in the line from the cap (lifting) end 1, with its pilot line from the other cylinder line so that it opens to lower the load 1)

(6)

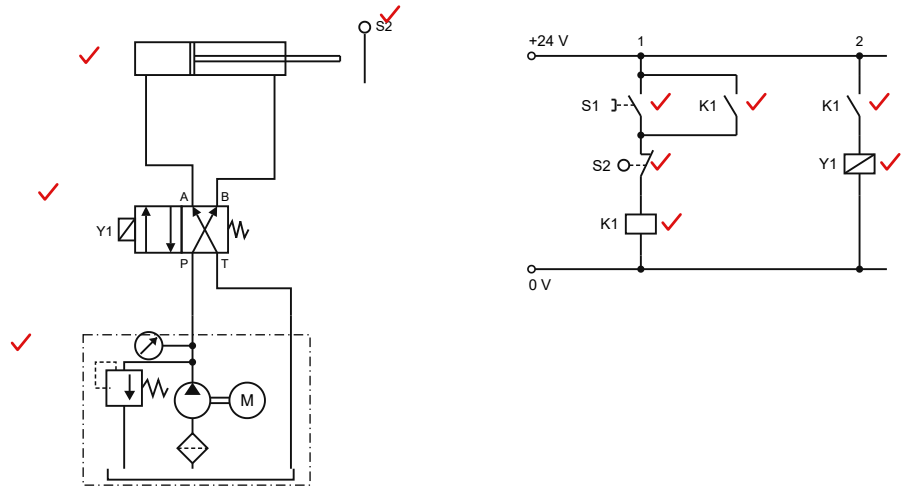
[20]

QUESTION 4

4.1 When the coil is energised, the electromagnet pulls the plunger / armature, which shifts the valve spool ✓ and changes the direction of flow of the oil ✓. (2)

4.2 An NO push button's contacts are open until it is pressed (it closes the circuit when pressed) ✓ – e.g. START ✓. An NC push button's contacts are closed until it is pressed (it opens the circuit when pressed) ✓ – e.g. STOP or emergency stop ✓. (4)

4.3



(Working circuit: power pack 1; 4/2 single-solenoid valve Y1 with spring return 1; double-acting cylinder correctly connected 1; limit switch S2 at the end of the stroke 1. Control circuit: +24 V and 0 V lines 1; S1 (NO) 1 with K1 holding contact in parallel 1; S2 (NC) in series with the K1 coil 1; K1 coil 1; separate rung with a K1 contact switching Y1 1)

(10)

4.4 Any FOUR ✓✓✓✓: switch off the pump and lock out / isolate the supply before working; release (bleed) all pressure from the system and accumulators; lower or support suspended loads first; never feel for leaks with your hand (oil jets can penetrate skin); wear safety glasses and gloves; use only parts and hoses rated for the system pressure; clean up spilled oil immediately. (4)

[20]

QUESTION 5

5.1 In order ✓✓✓✓✓: collect information about the fault (talk to the operator, look at the symptoms); study the circuit diagrams and how the system should work; identify the possible causes / isolate the faulty section; test and measure to find the faulty component; repair or replace it and test the whole system; record the fault and the repair. (5)

5.2 Any THREE causes with remedies, ✓✓ each: worn pump (low flow) – repair or replace the pump; leaking piston seals – replace the seals; flow control valve set too low – adjust it; relief valve set too low or stuck open – adjust / repair it; low oil level or blocked suction filter – top up oil / clean or replace the filter; internal leakage in the directional valve – repair or replace the valve. (6)

5.3 Set the meter to the correct DC (or AC) voltage range, higher than the expected voltage ✓; red lead in the V socket and black lead in COM ✓; with the circuit switched on, hold the probes across the two terminals of the coil (in parallel) without touching the metal tips ✓; read the value, then remove the probes and switch the meter off ✓. (4)

5.4 Hot oil becomes thin (lower viscosity), so leakage increases and lubrication is poorer ✓; seals and hoses are damaged and the oil breaks down (oxidises) faster ✓; components wear quickly and the system loses efficiency ✓. Any TWO causes ✓✓: relief valve opening continuously (set too low); oil level too low; blocked cooler / dirty oil; worn pump with internal leakage; wrong oil viscosity; system working at too high a pressure for too long. (5)

[20]

TOTAL: 100

END OF MARKING GUIDELINE